

A Statistical Physics of Language Model Reasoning

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Abstract

Transformer LMs show emergent reasoning that resists mechanistic understanding. We offer a statistical physics framework for continuous-time chain-of-thought reasoning dynamics. We model sentence-level hidden state trajectories as a stochastic dynamical system on a lower-dimensional manifold. This drift-diffusion system uses latent regime switching to capture diverse reasoning phases, including misaligned states or failures. Empirical trajectories (8 models, 7 benchmarks) show a rank-40 projection (balancing variance capture and feasibility) explains 50% variance. We find four latent reasoning regimes. An SLDS model is formulated and validated to capture these features. The framework enables low-cost reasoning simulation, offering tools to study and predict critical transitions like misaligned states or other LM failures.

1. Introduction

Transformer LMs (Vaswani et al., 2017), trained for next-token prediction (Radford et al., 2019; Brown et al., 2020), show emergent reasoning like complex cognition (Wei et al., 2022). Standard analyses of discrete components (e.g., attention heads (Elhage et al., 2021; Olsson et al., 2022)) provide limited insight into longer-scale semantic transitions in multi-step reasoning (Allen-Zhu & Li, 2023; López-Otal et al., 2024). Understanding these high-dimensional, prediction-shaped semantic trajectories, particularly how they might cause misaligned states, is a key challenge (Li et al., 2023; Nanda et al., 2023).

We model reasoning as a continuous-time dynamical system, drawing from statistical physics (Chaudhuri & Fiete, 2016; Schuecker et al., 2018). Sentence-level hidden states $h(t) \in \mathbb{R}^D$ evolve via a stochastic differential equation (SDE):

$$dh(t) = \mu(h(t), Z(t)) dt + B(h(t), Z(t)) dW(t), \quad (1)$$

with drift μ , diffusion B , Wiener process $W(t)$, and latent

regimes $Z(t)$. This decomposes trajectories into trends and variations, helping identify deviations. As full high-dimensional SDE analysis (e.g., $D > 2048$ for most LMs) is impractical, we use a lower-dimensional manifold capturing significant variance for modeling.

This continuous-time dynamical systems perspective offers several benefits:

Core Advantages

- **Principled Abstraction:** Enables a mathematically grounded, semantic-level view of reasoning, akin to statistical physics approximations, moving beyond token mechanics for robust interpretation of reasoning pathways and potential misalignments.
- **Tractable Latent Structure ID:** Makes analysis of reasoning trajectories feasible by focusing on a low-dimensional manifold (e.g., rank-40 PCA capturing 50% variance) that describes significant structured evolution.
- **Reasoning Regime Discovery:** Uncovers distinct latent semantic regimes with unique drift/variance profiles, suggesting context-driven switching and offering insight into how models might slip into different reasoning states (Appx. E).
- **Efficient Surrogate Model:** Our SLDS accurately models and reconstructs reasoning trajectories with significant computational savings, facilitating the study of how reasoning processes unfold.
- **Failure Mode Analysis:** Provides tools to study critical transitions, robustness, and predict inference-time failure modes or misaligned states in LLM reasoning.

Chain-of-thought (CoT) prompting (Wei et al., 2022; Wang et al., 2023) has demonstrated that LMs can follow structured reasoning pathways, hinting at underlying processes amenable to a dynamical systems description. While prior

work has applied continuous-time models to neural dynamics generally, the explicit modeling of transformer reasoning at these semantic timescales, particularly as an approximation for impractical full-dimensional analysis, has been largely unexplored. Our work bridges this gap by pursuing an SDE-based perspective informed by empirical analysis of transformer hidden-state trajectories.

This paper is structured as follows: Section 2 introduces the mathematical formalism of SDEs and regime switching. Section 3 details our data collection and initial empirical findings that motivate the model, including the practical need for dimensionality reduction. Section 4 formally defines the SLDS model. Section 5 presents experimental validation, including model fitting, generalization, ablation studies, and a case study on modeling adversarial belief shifts as an example of predicting misaligned states.

2. Mathematical Preliminaries

We conceptualize the internal reasoning process of a transformer LM as a continuous-time stochastic trajectory evolving within its hidden-state space. Let $h_t \in \mathbb{R}^D$ be the final-layer residual embedding extracted at discrete sentence boundaries $t = 0, 1, 2, \dots$. To capture the rich semantic evolution across reasoning steps, we treat these discrete embeddings as observations of an underlying continuous-time process $h(t) : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}^D$. The direct analysis of such a process in its full dimensionality (e.g., $D \geq 2048$) is often computationally prohibitive. We therefore aim to approximate its dynamics using SDEs, potentially in a reduced-dimensional space.

Definition 2.1 (Itô SDE). An Itô stochastic differential equation on the state space $\mathbb{R}^D$ is given by:

$$dh(t) = \mu(h(t))dt + B(h(t))dW(t), \quad h(0) \sim p_0, \quad (2)$$

where $\mu : \mathbb{R}^D \rightarrow \mathbb{R}^D$ is the deterministic *drift* term, encoding persistent directional dynamics. The matrix $B : \mathbb{R}^D \rightarrow \mathbb{R}^{D \times D'}$ is the *diffusion* term, modulating instantaneous stochastic fluctuations. $W(t)$ is a D' -dimensional Wiener process (standard Brownian motion), and p_0 is the initial distribution. The noise dimension D' can be less than or equal to the state dimension D .

The drift $\mu(h(t))$ represents systematic semantic or cognitive tendencies, while the diffusion $B(h(t))$ accounts for fluctuations due to local uncertainties, token-level variations, or inherent model stochasticity. Standard conditions ensure the well-posedness of such SDEs:

Theorem 2.1 (Well-Posedness (Øksendal, 2003)). *If μ and B satisfy standard Lipschitz continuity and linear growth conditions (see Appendix A), the SDE*

$$dh(t) = \mu(h(t))dt + B(h(t))dW(t) \quad (3)$$

has a unique strong solution for a given D' -dimensional Wiener process $W(t)$.

We focus on dynamics at the sentence level:

Definition 2.2 (Sentence-Stride Process). The *sentence-stride* hidden-state process is the discrete sequence $\{h_t\}_{t \in \mathbb{N}}$ obtained by extracting the final-layer transformer state immediately following each detected sentence boundary. This emphasizes mesoscopic, semantic-level changes over finer-grained token-level variations.

To analyze these dynamics in a computationally manageable way, particularly given the high dimensionality D of $h(t)$, we utilize projection-based dimensionality reduction. The goal is to find a lower-dimensional subspace where the most significant dynamics, for the purpose of modeling the SDE, unfold.

Definition 2.3 (Projection Leakage). Given an orthonormal matrix $V_k \in \mathbb{R}^{D \times k}$ (where $V_k^\top V_k = I_k$), the *leakage* of the drift μ under perturbations v orthogonal to the image of V_k (i.e., $v \perp \text{Im}(V_k)$) is

$$L_k = \sup_{\substack{x \in \mathbb{R}^D, \|v\| \leq \epsilon \\ v^\top V_k = 0}} \frac{\|\mu(x+v) - \mu(x)\|}{\|\mu(x)\|}.$$

A small leakage L_k implies that the drift’s behavior relative to its current direction is not excessively altered by components outside the subspace spanned by V_k , making the subspace a reasonable domain for approximation.

Assumption 2.1 (Approximate Projection Closure for Modeling). For practical modeling of the SDE (Eq. 2), we assume there exists a rank k (e.g., $k = 40$ in our work, chosen based on empirical variance and computational trade-offs) and a perturbation scale $\epsilon > 0$ such that $L_k \ll 1$. This allows the approximation of the drift within this k -dimensional subspace:

$$\mu(h(t)) \approx V_k V_k^\top \mu(h(t))$$

holds up to an error of order $O(L_k)$. This assumption underpins the feasibility of our low-dimensional modeling approach, enabling the analytical treatment inspired by statistical physics.

Empirical observations of reasoning trajectories suggest abrupt shifts, potentially indicating transitions between different phases of reasoning or slips into misaligned states. This motivates a regime-switching framework:

Definition 2.4 (Regime-Switching SDE). Let $Z(t) \in \{1, \dots, K\}$ be a latent continuous-time Markov chain with a transition rate matrix $T \in \mathbb{R}^{K \times K}$. The corresponding regime-switching Itô SDE is:

$$dh(t) = \mu_{Z(t)}(h(t))dt + B_{Z(t)}(h(t))dW(t), \quad (4)$$

where each latent regime $i \in \{1, \dots, K\}$ has distinct drift μ_i and diffusion B_i functions. This allows for context-dependent dynamic structures (Ghahramani & Hinton, 2000), crucial for capturing diverse reasoning pathways.

These definitions establish the mathematical foundation for our analysis of transformer reasoning dynamics as a tractable approximation of a more complex high-dimensional process.

3. Data and Empirical Motivation

We build a corpus of sentence-aligned hidden-state trajectories from transformer-generated reasoning chains across a suite of models (Mistral-7B-Instruct (Jiang et al., 2023), Phi-3-Medium (Abdin et al., 2024), DeepSeek-67B (DeepSeek-AI et al., 2024), Llama-2-70B (Touvron et al., 2023), Gemma-2B-IT (Gemma Team & Google DeepMind, 2024), Qwen1.5-7B-Chat (Bai et al., 2023), Gemma-7B-IT (also (Gemma Team & Google DeepMind, 2024)), Llama-2-13B-Chat-HF (also (Touvron et al., 2023))) and datasets (StrategyQA (Geva et al., 2021), GSM-8K (Cobbe et al., 2021), TruthfulQA (Lin et al., 2022), BoolQ (Clark et al., 2019), OpenBookQA (Mihaylov et al., 2018), HellaSwag (Zellers et al., 2019), PiQA (Bisk et al., 2020), CommonsenseQA (Talmor et al., 2021; 2019)), yielding roughly 9,800 distinct trajectories spanning $\sim 40,000$ sentence-to-sentence transitions.

3.1. Sentence-Level Dynamics and Manifold Structure for Tractable Modeling

First, we confirmed that sentence-level increments effectively capture semantic evolution. Figure 1(a) compares the cumulative distribution functions (CDFs) of jump norms ($\|\Delta h_t\|$) at both token and sentence strides. Token-level increments show a noisy distribution skewed towards small values, primarily reflecting syntactic variations. In contrast, sentence-level increments are orders of magnitude larger, clearly indicating significant semantic shifts and validating our choice of sentence-stride analysis. To reduce "jitter" from minor variations, we filtered out transitions below a minimum threshold ($\|\Delta h_t\| \leq 10$ in normalized units), yielding cleaner semantic trajectories.

To uncover underlying geometric structures that could make modeling tractable, we applied Principal Component Analysis (PCA) (Jolliffe, 2002) to the sentence-stride embeddings. We found that a relatively low-dimensional projection (rank $k = 40$) captures approximately 50% of the total variance in these reasoning trajectories (details in Appendix A). While reasoning dynamics occur in a high-dimensional embedding space, this finding suggests that a significant portion of their variance is concentrated in a lower-dimensional subspace. This is crucial because constructing and analyzing a

stochastic process (like a random walk or SDE) in the full embedding dimension (e.g., 2048) is often impractical. The rank-40 manifold thus provides a computationally feasible domain for our dynamical systems modeling, not necessarily because the process is strictly confined to it, but because it offers a practical and informative approximation.

3.2. Linear Predictability and Multimodal Residuals

To assess the predictive structure of the semantic drift within this tractable manifold, we performed a global ridge regression (Hoerl & Kennard, 1970), fitting a linear model to predict subsequent sentence embeddings from previous ones:

$$h_{t+1} \approx Ah_t + c, \quad (5)$$

$$(A, c) = \arg \min_{A, c} \sum_t \|\Delta h_t - (A - I)h_t - c\|^2 + \lambda \|A\|_F^2. \quad (6)$$

Using a modest regularization ($\lambda = 1.0$), this global linear model achieved an $R^2 \approx 0.51$, indicating substantial linear predictability in sentence-to-sentence transitions.

However, an examination of the residuals from this linear fit, $\xi_t = \Delta h_t - [(A - I)h_t + c]$, revealed persistent multimodal structure, even after the linear drift component was removed (Figure 1(b)). This multimodality suggests the presence of distinct underlying dynamic states or phases—some potentially representing "misaligned states" or divergent reasoning paths—that are not captured by a single linear model.

Inspired by Langevin dynamics, where a particle in a multi-well potential $U(x)$ can exhibit metastable states (Appendix E), we interpret these multimodal residual clusters as evidence of distinct latent reasoning regimes. The stationary probability distribution $p_{st}(x) \propto e^{-U(x)/D}$ for an SDE $dx = -U'(x)dt + \sqrt{2D}dW_t$ becomes multimodal if $U(x)$ has multiple minima and noise D is sufficiently low. Analogously, the observed clusters in our residual analysis point towards the existence of multiple metastable semantic basins in the reasoning process. This strongly motivates the introduction of a latent regime structure to adequately model these richer, nonlinear dynamics and to understand how an LLM might transition between effective reasoning and potential failure modes.

4. A Switching Linear Dynamical System for Reasoning

The empirical evidence that a significant portion of variance is captured by a low-dimensional manifold (making it a practical subspace for analysis, as directly modeling a 2048-dim random walk is often infeasible) and the observation of multimodal residuals motivate a model that combines linear dynamics within distinct regimes with switches between